MATH-361 Probability and Statistics

Measures of Central Tendency

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Measure of Central Tendency

1. Mean $\bar{X} = \frac{\sum X}{n}$
2. Median $MD = \begin{cases} X_{\frac{n+1}{2}} & \text{if } n \text{ is odd} \\ \frac{1}{2}\left(X_{\frac{n}{2}} + X_{(\frac{n}{2}+1)}\right) & \text{if } n \text{ is even} \end{cases}$
3. Mode

Measure of Variation

1. Range
2. Variance/Standard Deviation

Statistic

A statistic is a characteristic or measure obtained by using the **data values from a sample**.

Parameter

A parameter is a characteristic or measure obtained by using **all the data values** from a specific population.

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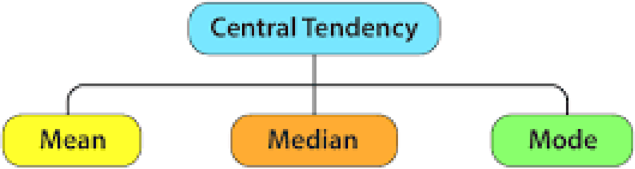
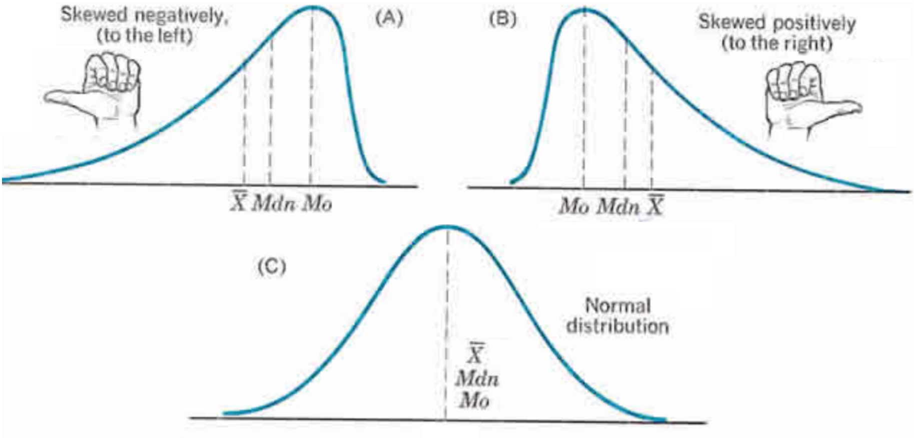



Skewed negatively,
(to the left)
(A)
$\bar{X}$ Mdn Mo
(B)
Skewed positively
(to the right)
Mo Mdn $\bar{X}$
(C)
Normal
distribution
$\bar{X}$
Mdn
Mo

Mean $\bar{X} = \frac{X_1+X_2+\cdots+X_n}{n}$

The mean, also known as the arithmetic average, is found by adding the values of the data and dividing by the total number of values.

For example, the mean of 3, 2, 6, 5, and 4 is found by adding

$$3+2+6+5+4=20 \quad \text{and dividing by 5}$$

hence, the mean of the data is $20/5 = 4$.

Mean $\bar{X} = \frac{X_1+X_2+\cdots+X_n}{n}$

The values of the data are represented by $X's$.

In this data set 3, 2, 6, 5, and 4

$$X_1 = 3,\ X_2 = 2,\ X_3 = 6,\ X_4 = 5, \quad \text{and } X_5 = 4.$$

The symbol $\bar{X}$ represents the sample mean.

$$\bar{X} = \frac{X_1 + X_2 + \cdots + X_n}{n}, \quad n \text{ is the total number of values in the sample.}$$

To show a sum of the total X values, the symbol $\sum$, (the capital Greek letter sigma) is used, and $\sum X$ means to find the sum of the X values in the data set.

$$\boxed{\bar{X} = \frac{\sum X}{n}}$$

The median is the midpoint of the data array. The symbol for the median is **MD**.

1. Arrange the data in order.
2. Select the middle point.

$$MD = \begin{cases} X_{\frac{n+1}{2}} & \text{if } n \text{ is odd} \\ \frac{1}{2}\left(X_{\frac{n}{2}} + X_{(\frac{n}{2}+1)}\right) & \text{if } n \text{ is even} \end{cases}$$

The value that occurs most often in a data set is called the **mode**.

The **midrange** is defined as the sum of the lowest and highest values in the data set, divided by 2. The symbol **MR** is used for the midrange.

$$MR = \frac{\text{Highest value} + \text{Lowest value}}{2}$$

Find the weighted mean of a variable X by multiplying each value by its corresponding weight and dividing the sum of the products by the sum of the weights.

$$\frac{w_1X_1 + w_2X_2 + \cdots + w_nX_n}{w_1 + w_2 + \cdots + w_n}$$

Here $w_1, w_2, \ldots, w_n$ are the weights and $X_1, X_2, \cdots, X_n$ are the values.

A student received an A in English Composition I (3 credits), a C in Introduction to Psychology (3 credits), a B in Biology I (4 credits), and a D in Physical Education (2 credits). Assuming $A = 4$ grade points, $B = 3$ grade points, $C = 2$ grade points, $D = 1$ grade point, and $F = 0$ grade points, find the student's grade point average.

Course	Credits (w)	Grade (X)
English Composition I	3	A (4 points)
Introduction to Psychology	3	C (2 points)
Biology I	4	B (3 points)
Physical Education	2	D (1 point)

$$\bar{X} = \frac{\sum wX}{\sum w} = \frac{3 \cdot 4 + 3 \cdot 2 + 4 \cdot 3 + 2 \cdot 1}{3 + 3 + 4 + 2}$$

Summary of Measures of Central Tendency

Measure	Definition	Symbol(s)
Mean	Sum of values, divided by total number of values	μ, $\overline{X}$
Median	Middle point in data set that has been ordered	MD
Mode	Most frequent data value	None
Midrange	Lowest value plus highest value, divided by 2	MR

The following measurements were recorded for the drying time, in hours, of a certain brand of latex paint.

$$\begin{array}{cccccccc} 3.4 & 2.5 & 4.8 & 2.9 & 3.6 & 2.8 & 3.3 & 5.6 \\ & 3.7 & 2.8 & 4.4 & 4.0 & 5.2 & 3.0 & 4.8 \end{array}$$

Assume that the measurements are a simple random sample.

(a) What is the sample size for the above sample?

(b) Calculate the sample mean for these data.

(c) Calculate the sample median.

(d) Compute the 20% trimmed mean for the above data set.

(e) Is the sample mean for these data more or less descriptive as a center of location than the trimmed mean?

The **range** is the highest value minus the lowest value. The symbol R is used for the range.

$$R = \text{ highest value - lowest value}$$

The variance is the average of the squares of the distance each value is from the mean. The symbol for the population variance is σ^2(σ is the Greek lowercase letter sigma). The formula for the population variance is

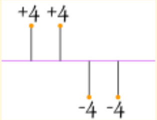
$$\sigma^2 = \frac{\sum(X-\mu)^2}{N}$$

where $X=$ individual value, μ is population mean, N is population size

The standard deviation is the square root of the variance. The symbol for the population standard deviation is σ. The corresponding formula for the population standard deviation is

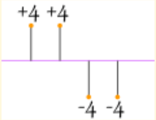
$$\sigma = \sqrt{\frac{\sum(X-\mu)^2}{N}}$$

If we just add up the differences from the mean ... the negatives cancel the positives:



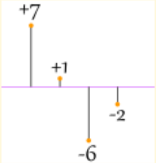
$$\frac{4+4-4-4}{4} = \boxed{0}$$

So that won't work. How about we use absolute values?



$$\frac{|4|+|4|+|-4|+|-4|}{4} = \frac{4+4+4+4}{4} = \boxed{4}$$

That looks good (and is the Mean Deviation), but what about this case:



$$\frac{|7|+|1|+|-6|+|-2|}{4} = \frac{7+1+6+2}{4} = \boxed{4}$$

Oh No! It also gives a value of 4, Even though the differences are more spread out.

A testing lab wishes to test two experimental brands of outdoor paint to see how long each will last before fading. The testing lab makes 6 gallons of each paint to test. Since different chemical agents are added to each group and only six cans are involved, these two groups constitute two small populations. The results (in months) are shown. Find the mean of each group.

Brand A	Brand B
10	35
60	45
50	30
30	35
40	40
20	25

Find the variance and standard deviation for the data set for brand A paint and brand B

$$\mu_A = 35$$

$$\sigma_A^2 = \frac{\sum (X - \mu_A)^2}{N}$$

$$= \frac{25^2 + 25^2 + 15^2 + 5^2 + 5^2 + 15^2}{6}$$

Variance $\sigma_A^2 = 291.7$

Standard Deviation $\sigma_A = 17.1$

$$\mu_B = 35$$

$$\sigma_B^2 = \frac{\sum (X - \mu_B)^2}{N}$$

$$= \frac{\cdots}{6}$$

Variance $\sigma_B^2 =$

Standard Deviation $\sigma_B =$

Sample Variance and Standard Deviation

When computing the variance for a sample, one might expect the following expression to be used

$$s^2 = \frac{\sum(X - \bar{X})^2}{n}, \qquad s = \sqrt{\frac{\sum(X - \bar{X})^2}{n}}$$

Sample variance and sample standard deviation are denoted by s^2 and s, respectively. Above formula does not give the best estimate of the population variance because when the population is large and the sample is small (usually less than 30), the variance computed by this formula usually underestimates the population variance. Therefore, instead of dividing by n, find the variance of the sample by dividing by $n-1$, giving a slightly larger value and an unbiased estimate of the population variance.

$$s^2 = \frac{\sum(X - \bar{X})^2}{n-1}, \qquad s = \sqrt{\frac{\sum(X - \bar{X})^2}{n-1}}$$

$$s^2 = \frac{\Sigma(X-\bar{X})^2}{n-1} = \frac{\Sigma(X^2 - 2\bar{X}\cdot X + \bar{X}^2)}{n-1}$$

$$s^2 = \frac{(\Sigma X^2) - \Sigma(2\bar{X}\cdot X) + \Sigma(\bar{X}^2)}{n-1} = \frac{(\Sigma X^2) - 2\bar{X}\cdot(\Sigma X) + \bar{X}^2\cdot(\Sigma 1)}{n-1}$$

$$s^2 = \frac{(\Sigma X^2) - 2\bar{X}\cdot(n\bar{X}) + \bar{X}^2\cdot n}{n-1} = \frac{(\Sigma X^2) - 2n\bar{X}^2 + n\bar{X}^2}{n-1}$$

$$s^2 = \frac{n(\Sigma X^2) - n^2\bar{X}^2}{n(n-1)} = \frac{n(\Sigma X^2) - (n\bar{X})^2}{n(n-1)}$$

Shortcut or Computational Formulas for s^2 and s

The shortcut formulas for computing the variance and standard deviation for data obtained from samples are as follows.

$$s^2 = \frac{n(\sum X^2) - (\sum X)^2}{n(n-1)}, \qquad s = \sqrt{\frac{n(\sum X^2) - (\sum X)^2}{n(n-1)}}$$

The Range Rule of Thumb

$$s \simeq \text{range}/4$$

Find the sample variance and standard deviation for the amount of European auto sales for a sample of 6 years shown. The data are in millions of dollars.

11.2, 11.9, 12.0, 12.8, 13.4, 14.3

Step-1 Find sum of all the values

$$\sum X = 11.2 + 11.9 + 12.0 + 12.8 + 13.4 + 14.3 = 75.6$$

Step-2 Square each value and find the sum.

$$\sum X^2 = 11.2^2 + 11.9^2 + 12.0^2 + 12.8^2 + 13.4^2 + 14.3^2 = 958.94$$

Step-3

$$s^2 = \frac{n(\sum X^2) - (\sum X)^2}{n(n-1)} = \frac{6(958.94) - (75.6)^2}{6 \times 5}$$

$$s^2 = 1.28, \qquad \text{and} \qquad s = 1.13$$

The three data sets have the same mean and range, but is the variation the same? Prove your answer by computing the standard deviation. Assume the data were obtained from samples.

a.	5,	7,	9,	11,	13,	15,	17
b.	5,	6,	7,	11,	15,	16,	17
c.	5,	5,	5,	11,	17,	17,	17

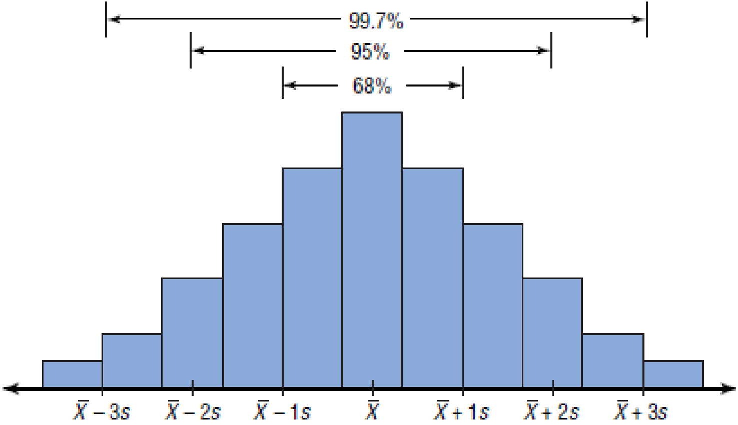
a. $s_a^2 = 18.67, \qquad s_a = 4.3$

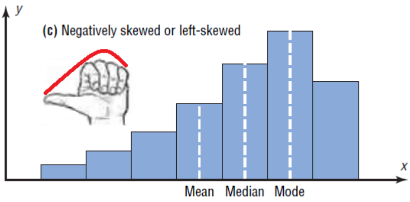
b. $s_b^2 = 25.67, \qquad s_b = 5.1$

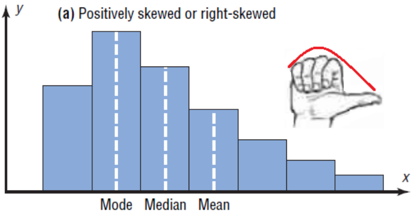
c. $s_c^2 = 36, \qquad s_c = 6$

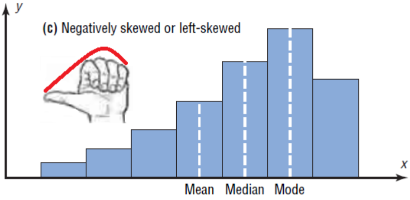
When a distribution is bell-shaped (or what is called normal), the following statements, which make up the empirical rule, are true. Approximately

1. 68% of data values will fall within 1 standard deviation of the mean.
2. 95% of data values will fall within 2 standard deviations of the mean.
3. 99.7% of data values will fall within 3 standard deviations of the mean.




(c) Negatively skewed or left-skewed
y
x
Mean
Median
Mode


(a) Positively skewed or right-skewed
y
x
Mode
Median
Mean


y
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x
Mean
Median
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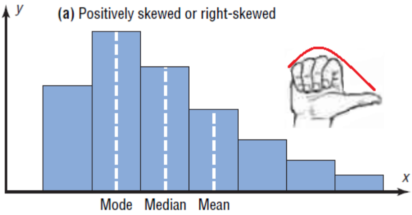
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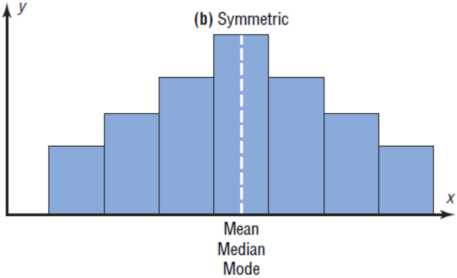
y
(b) Symmetric
x
Mean
Median
Mode

QUIZ

The mean of 25 observations is 36. The mean of first 13 observations is 32 and that of last 13 observations is 39. What is the value of 13^{th} observation?